

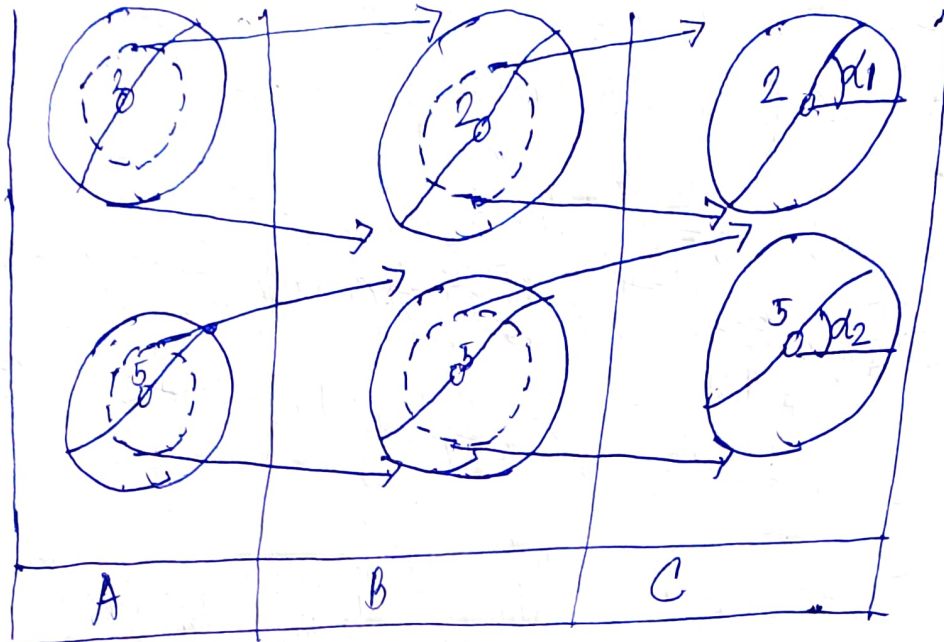


Fig. 32

Author: Let us turn to Fig. 32. Here in column A two portions of the graph of the function $s(t)$ are shown separately, namely, those in the vicinity of points 2 and 5 (in Fig. 31 these portions are identified by dash circles). In column B the portions of the graph close to points 2 and 5 are shown again, but this time with a two-fold increase in scale. Column C shows the result of another two-fold scale increase. Obviously, as the scale increases, the curvature of the graph $s(t)$ becomes less noticeable. We may say that the graph has a property of "linearity on a small scale.", which enables us to consider the slope of the graph at a specific point. In Fig 32 (in column C) it is shown that the slope of the graph at point 2 is d_1 (the slope is measured relative to the t -axis), while the slope at point 5 is d_2 , and clearly $d_2 < d_1$.

Denote the slope of the curve $s(t)$ at the moment t by $d(t)$. Then $\tan d(t)$ is said to be the rate of change of the function $s(t)$ at the moment t , or simply the instantaneous velocity.

Reader: But why tangent?

Author: You immediately come to it by considering portion 1-2 of the graph in Fig. 31. This portion represents a uniform motion of the body, the rate of change of $s(t)$ being identical at all points. Obviously, it equals the average velocity during the time interval $t_2 - t_1$, which is $\frac{s_2 - s_1}{t_2 - t_1} = \tan \alpha$.